UNIT 5 · 8 hours

Introduction to State-Variable Analysis



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
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Learning objectives

- Define state, state variables and state space
- Write the state and output equations
- Draw the state-space block diagram
- List advantages of the state-space approach over the transfer function

Syllabus context

Introduction to state variable analysis: modelling of physical systems using the state-space approach, advantages of state space over the transfer function approach, state-space model using the physical variable approach for SISO systems.

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Basic definitions

State

The minimum set of variables whose values at t0 plus the input for t ≥ t0 determine the future behaviour.

State variables x1...xn

The variables of that set; n = order of the system = number of independent energy storage elements.

State vector and space

$x = [x1 \dots xn]^T$; the n-dimensional space of x is the state space.

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State-space model (LTI, SISO)

$$\dot{x} = A\,x + B\,u$$

state equation

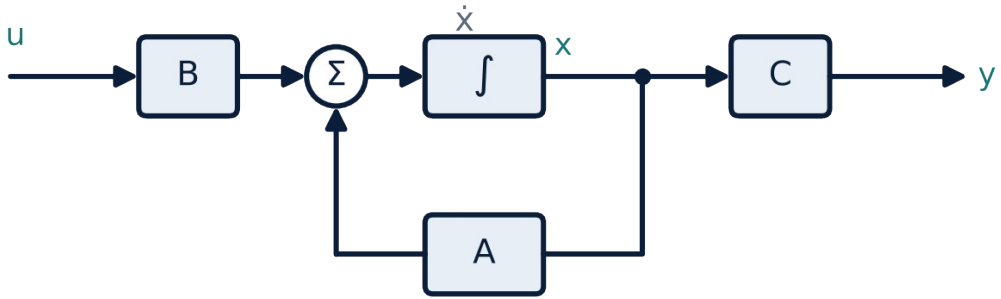
$$y = C\,x + D\,u$$

output equation

- x: n×1 state; u: input; y: output.
- A: n×n system matrix; B: n×1 input matrix; C: 1×n output matrix; D: direct transmission (often 0).
- Transfer function: $G(s) = C\,(sI - A)^{-1}\,B + D$; poles = eigenvalues of A ($\det(sI - A) = 0$).

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State-space block diagram



- Integrators hold the states.
- A: state feedback; B: input matrix; C: output matrix.
- Each state variable is the output of an integrator.

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State space versus transfer function

Transfer function

- Input–output only
- Zero initial conditions
- Single-input single-output (mainly)
- Linear time-invariant systems only

State space

- Describes internal states
- Includes initial conditions
- Multi-input multi-output easily
- Time-varying and non-linear systems
- Suited to computers and modern/optimal control

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Advantages of state-space analysis

- Handles multiple inputs and outputs naturally (MIMO).
- Includes initial conditions and internal behaviour.
- Applicable to non-linear and time-varying systems (after linearisation).
- Convenient for computer simulation (MATLAB ss, lsim).
- Foundation for state feedback, observers and optimal control.
- Controllability and observability tests possible.

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Key takeaways

- State: minimum set of variables describing the system's condition
- $\dot{x} = Ax + Bu,\ \ y = Cx + Du$
- $G(s) = C(sI - A)^{-1}B + D$; poles = eigenvalues of A
- Advantages: MIMO, initial conditions, nonlinear/time-varying, computer friendly

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Practice questions

Q1. Define state, state variable and state vector.

Q2. Write the standard state and output equations.

Q3. List four advantages of state space over transfer function.

Q4. How are poles related to A?